

Assignment 3: Type Checker for RustY

IC526 Advanced Compilers
Fall 2026

Implement the RustY type checker described in Lectures 8 and 9.

1. Your Task

- Complete the TODO functions in `lib/typecheck.ml`.
- Annotate all expressions using the typed AST in `lib/ast.ml`.
- Raise `TypeError` for invalid programs; error messages need not match exactly.

Required interface:

```
val tc_prog : Ast.prog -> Ast.tprog
```

Only `lib/typecheck.ml` will be graded. Helper functions may be added in this file.

2. Specification

- Names: function names must be unique. Parameter names must be unique within each function. Collect all signatures before checking bodies to support forward calls and recursion.
- Numeric literals: synthesis chooses `i32`; checking uses the expected `i32` or `usize`. There is no implicit conversion between these types.
- Operators: binary arithmetic checking propagates the expected numeric type to both operands. Comparisons synthesize both operands and require matching numeric types; boolean equality is unsupported. Unary `-` requires an `i32` operand; `!`, `&&`, and `||` require `bool` operands.
- Arrays: check element types and lengths. Empty arrays need an expected type. Check indices against `usize`.
- Scope: check initializers before adding bindings. Allow shadowing; declarations inside `if` branches and `while` bodies do not escape.
- Mutability: assignments require a mutable location. Parameter bindings are immutable; writing through an `&mut` parameter is allowed.
- Calls and statements: reject undefined names, wrong argument counts, and type mismatches. Conditions require `bool`; returns must match the declared return type. `println!` accepts only numeric values.
- Borrowing: check reference arguments and forwarding against parameter types. Among the reference arguments of each call, shared borrows may repeat, but an exclusive borrow conflicts with any other borrow of the same variable.

A `main` function and an analysis proving that every path returns are not required.

3. Testing

From the project root, with the course opam switch selected:

```
dune build
dune exec rustyc -- --typecheck input.rs
```